

PHYSICS 7B

UC BERKELEY

SPRING 2026

7B Discussion Problems, Week 12

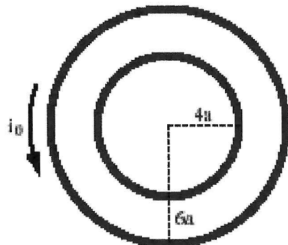
Topics

Ampere's Law and the Biot-Savart Law

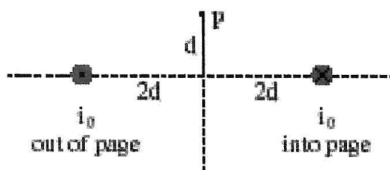
Week 12 Discussion Problems

Ampere's Law and the Biot-Savart Law

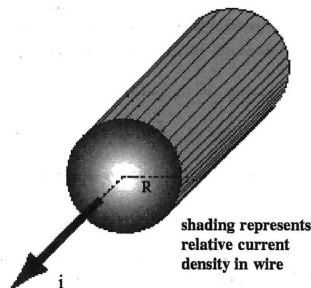
- Two circular wire loops are fixed in place as shown. The large loop carries current i_0 counter-clockwise as viewed from above.



- Suppose we want the magnetic field to vanish at the common center of the loops. Should we generate a clockwise current in the small loop or a counter-clockwise current?
 - How strong a current will we need to generate? (Hint: At the center of a circular current loop of radius R the loop's magnetic field has magnitude $B_{\text{loop}} = \frac{\mu_0 I}{2R}$.)
- Two straight wires are fixed in place near one another, and carry equal currents i_0 in opposite directions.



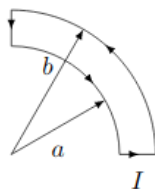
- Sketch the two contributions to the net magnetic field at point P.
 - Add these vectors graphically to obtain a sketch of the net magnetic field vector at P.
 - Calculate the magnitude and direction of the magnetic field at P.
- A thick, infinite wire of radius R carries total current i distributed *non-uniformly* across its cross section. The current density within the wire (in amps per square meter) is given by $j(r) = j_0(r/R)^4$.



- Sketch the magnetic field created by the wire, at points both inside and outside the wire.
- Find the strength of the magnetic field created by the wire, at any point inside the wire ($r < R$).
- Find the strength of the magnetic field created by the wire, at any point outside the wire ($r > R$).

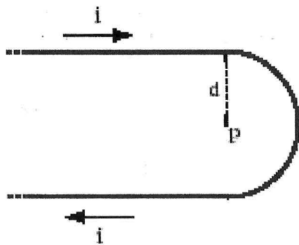
4. Problem:

Problem 20.4: (Magnetic Field from a Quarter Circle) Suppose there is a wire in the shape shown below with a steady current I travelling clockwise. What is the magnetic field at point P ?



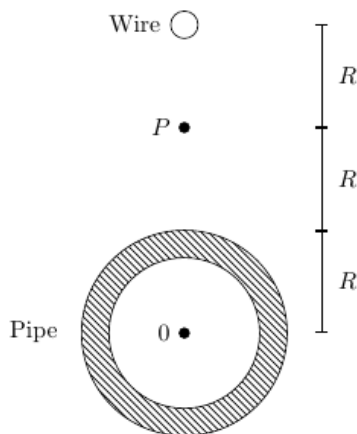
5. A long wire is bent into the hairpin-like shape shown in the figure.

- What is the direction of the magnetic field at the indicated point P , which lies at the center of the half-circle?
- What is the magnitude of the magnetic field at that point? (Hint: At the center of a circular current loop of radius R the loop's magnetic field has magnitude $B_{\text{loop}} = \frac{\mu_0 I}{2R}$.)
- Suppose an electron is at rest at point P . What force does the magnetic field exert on it?



6. Problem:

Problem 19.5: In the figure below, a long circular pipe with outside radius R carries a (uniformly distributed) current I into the page. A wire runs parallel to the pipe at a distance of $3R$ from center to center. Find the magnitude and direction (into or out of the page) of the current in the wire such that the net magnetic field at point P has the same magnitude as the net magnetic field at the center of the pipe but is in the opposite direction.



Week 12 Solutions

Ampere's Law and the Biot-Savart Law

1. (a) CW
(b)

$$\frac{\mu_0 I}{2 \times 6a} - \frac{\mu_0 I}{2 \times 4a} = 0 \quad (1)$$

$$\frac{I_0}{6} - \frac{I}{4} = 0 \quad (2)$$

$$I = \frac{2I_0}{3} \quad (3)$$

2. (a) Sketch. Magnetic field due to left wire points upper left and magnetic field due to right wire points upper right.
(b) The horizontal components cancel out, and the net magnetic field points straight up.
(c) We know the magnetic field around a straight wire is $B(r) = \frac{\mu_0 I}{2\pi r}$. Here $r = \sqrt{5}d$. Taking the vertical component only, we have

$$B = 2 \cdot \frac{\mu_0 I}{2\pi\sqrt{5}d} \frac{1}{\sqrt{5}} = \frac{\mu_0 I}{5\pi d}. \quad (4)$$

3. (a) Sketch. Field lines are concentric circles, in the ccw direction.
(b) Take a circle of radius $r < R$ as the Amperian loop. The current enclosed in this loop is given by

$$I_{\text{enc}} = \int_0^r j(r) 2\pi r dr = \frac{2\pi j_0}{R^4} \int_0^r r^5 dr = \frac{\pi j_0 r^6}{3R^4}. \quad (5)$$

Then, by Ampere's law, $B(r)2\pi r = \mu_0 I_{\text{enc}}$, we have

$$B(r) = \frac{\mu_0 r^5 j_0}{6R^4}. \quad (6)$$

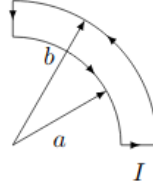
- (c) Take a circle of radius $r > R$ as the Amperian loop. The current enclosed in this loop is now

$$I_{\text{enc}} = \int_0^R j(r) 2\pi r dr = \frac{2\pi j_0}{R^4} \int_0^R r^5 dr = \frac{\pi j_0 R^2}{3}. \quad (7)$$

Again use Ampere's law and we have

$$B(r) = \frac{\mu_0 R^2 j_0}{6\pi r}. \quad (8)$$

4. Solution:

Solution 20.4:

This can also be done re-using old problems. The two straight sides of the wire are parallel or anti-parallel to the $\hat{r}$ vector, so the cross-products vanish there. They therefore do not contribute to the magnetic field. Using the right-hand rule, both semi-circles contribute a component in the vertical direction. To find the magnitudes of those components, it is easiest to find the magnetic field due to a quarter circle of arbitrary radius and current: $\mathbf{B}_{\text{Q.C.}}(R, I)$.

However, a quarter circle is just a quarter of a loop, so we can just use dB_z from above, with $z = 0$. Therefore

$$\mathbf{B}_{\text{Q.C.}}(R, I) = \int_0^{\pi/2} \frac{\mu_0 I}{4\pi} \frac{R^2}{(R^2 + 0^2)^{3/2}} d\theta \hat{z} = \frac{\mu_0 I}{8R} \hat{z}.$$

The total magnetic field is then:

$$\mathbf{B} = \mathbf{B}_{\text{Q.C.}}(a, -I) + \mathbf{B}_{\text{Q.C.}}(b, I) = \frac{\mu_0 I}{8} \left(\frac{1}{b} - \frac{1}{a} \right) \hat{z}$$

5. Let us get the B-field of a semi-infinite wire. Let's set our coordinate system so $+\hat{x}$ points to the right, $+\hat{y}$ points up the page, and $+\hat{z}$ points out of the page.

$$d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{d\vec{l} \times \hat{r}}{r^2} \quad (9)$$

$$dB = \frac{\mu_0 I}{4\pi} \frac{dx \sin\theta}{r^2} \quad (10)$$

$$= \frac{\mu_0 I}{4\pi} \frac{1}{x^2 + y^2} \frac{y}{\sqrt{x^2 + y^2}} dx. \quad (11)$$

Therefore,

$$B = \frac{\mu_0 I}{4\pi} \int_{-\infty}^0 \frac{y}{(x^2 + y^2)^2} dx \quad (12)$$

$$\vec{B} = \frac{\mu_0 I}{4\pi} \frac{1}{y} (-\hat{z}) \quad (13)$$

Then the total magnetic field is

$$\vec{B}_z = -\hat{z} \left(\frac{\mu_0 I}{4\pi d} + \frac{\mu_0 I}{4\pi d} + \frac{\mu_0 I}{4d} \right) \quad (14)$$

$$= -\frac{\mu_0 I}{4\pi} \left(1 + \frac{2}{\pi} \right) \hat{z} \quad (15)$$

$$(16)$$

- (a) Into the page ($-\hat{z}$)

- (b) Given above.
 (c) 0, the particle isn't moving!

6. Solution:

Solution 19.5: External to the wire the magnetic field of the pipe is indistinguishable from that of a wire, so

$$\mathbf{B}_{\text{pipe}}(r > R) = -\frac{\mu_0 I}{2\pi r} \hat{\theta}.$$

The minus sign is because the current is into the page. So in particular $\mathbf{B}(P) = \frac{\mu_0 I}{2\pi(2R)} \hat{x}$ where I have taken $\hat{x}$ to point to the right in the picture. At the center of the pipe, which I will call 0, there is no magnetic field from the pipe by Ampère's Law.

Meanwhile the field from the wire at P is

$$\mathbf{B}_{\text{wire}}(P) = \frac{\mu_0 I_w}{2\pi R} \hat{x}$$

and at the origin is

$$\mathbf{B}_{\text{wire}}(O) = \frac{\mu_0 I_w}{2\pi(3R)} \hat{x}.$$

Here I have taken I_w to be the (possibly negative) strength of the current coming out of the page in the wire. We want to satisfy the condition $\mathbf{B}(P) = -\mathbf{B}(O)$. So:

$$\mathbf{B}(P) = \mathbf{B}_{\text{pipe}}(P) + \mathbf{B}_{\text{wire}}(P) = \frac{\mu_0 I}{2\pi(2R)} \hat{x} + \frac{\mu_0 I_w}{2\pi R} \hat{x}$$

and

$$\mathbf{B}(O) = \mathbf{B}_{\text{wire}}(O) = \frac{\mu_0 I_w}{2\pi(3R)} \hat{x}.$$

So our condition means that

$$\frac{\mu_0 I}{2\pi(2R)} + \frac{\mu_0 I_w}{2\pi R} = -\frac{\mu_0 I_w}{2\pi(3R)}.$$

Dividing both sides by $\frac{\mu_0}{2\pi R}$,

$$\frac{I}{2} + I_w = -\frac{I_w}{3} \implies I_w = -\frac{3}{8}I.$$